SDC HACKATHON 2026 · DOMAIN 1 · FUTURE-READY WORKFORCE

Workforce Futures

From who you are today to where you can be in the AI era.



[MK GreenCode] · https://workforce-future.replit.app/

WHAT LIVE DATA SHOWS

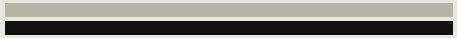
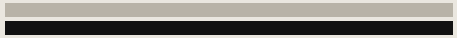
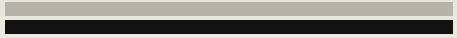
In three months, demand moved both ways.

Share of job posts asking for each skill: May–Jul Aug–Oct 2026

▲ Rising

Digital marketing & analytics		1.5% → **2.3%**
Cybersecurity		7.4% → **9.7%**
Databases & SQL		13.4% → **16.4%**

▼ Cooling

Data visualisation & reporting		1.9% → **1.1%**
Spreadsheets & data handling		2.3% → **1.6%**
Sustainability & ESG reporting		1.3% → **0.9%**

MEANWHILE, AI KEEPS MOVING

1,005 → 1,232
new research papers on LLM agents per three months (arXiv)

11.4% → 13.8%
of job posts now mention AI agents

19 → 28
job posts naming MCP, a new standard for linking AI agents to tools

Hacker News "Who is hiring?": 1,452 posts (May–Jul) and 1,158 posts (Aug–Oct 2026); arXiv abstracts for the same windows. Snapshot of 22 Oct 2026, fetched by the app.

01 · THE PROBLEM · ONE PERSON, ONE MOMENT OF NEED

Meet Mei Ling, 45.

“I key in delivery orders, prepare invoices in Excel and answer customers in three languages. Is my job going away, and what should I learn next?”

Office clerk at a logistics firm in Shah Alam · 18 years · 5 hours a week to learn · free courses only

WHY IT IS HARD FOR HER TODAY

The shift is invisible. Her data-entry and spreadsheet work is what job posts ask for less.

Advice is generic. “Learn AI”, but which skill, for which job, and how long?

Learning is scattered. Dozens of course sites, unclear prices, none tied to a next job.

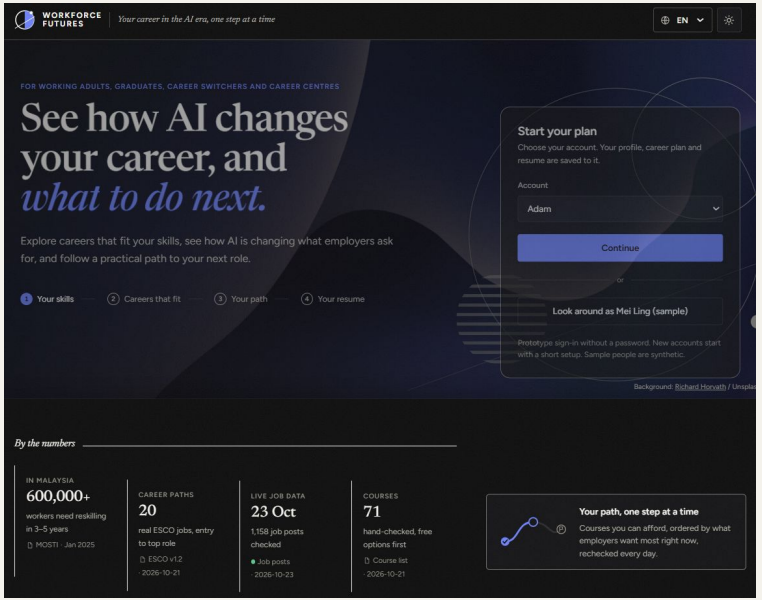
Her manager plans in a spreadsheet, with no shared view of the team’s gaps.

> “How might we build an AI-powered solution that **continuously tracks AI advancement** to predict which skills and jobs will emerge or decline, **matches** today’s workforce and tomorrow’s talent to the right future roles, and generates **personalised reskilling pathways**, while showing how each role and skill connects to the **17 UN Sustainable Development Goals**?”

02 · OUR SOLUTION

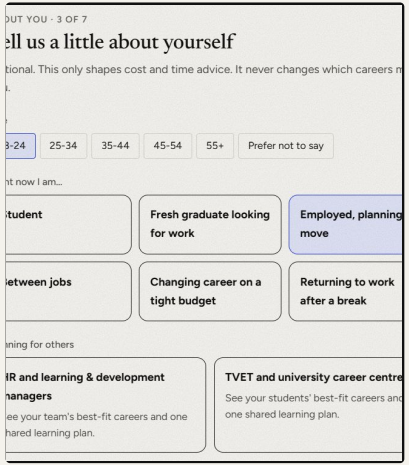
So Workforce Futures is Introduced

Workforce Futures turns public trend signals into a personal, explainable plan. It fetches job posts and AI research every day, measures which skills are rising or cooling, scores 20 future careers for the person with an open formula (skills, demand, transferable skills and interests, minus AI exposure from the ILO index), and builds an ordered reskilling path with checked courses that fit their budget and weekly hours. Every role and skill is linked to official UN SDG targets with a reason. Large language model is plugged in and only reads CVs and rewords explanations, under guardrails. It never chooses careers, courses or scores. Instead of deciding everything, the app let user to decide their future themselves



FOR PERSONAL JOB SEARCHER

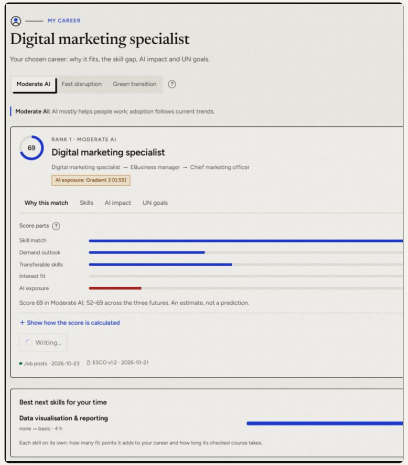
Tailored Pathway with 4 Easy Steps



1 Tell us about yourself



2 See the Future Skills Radar



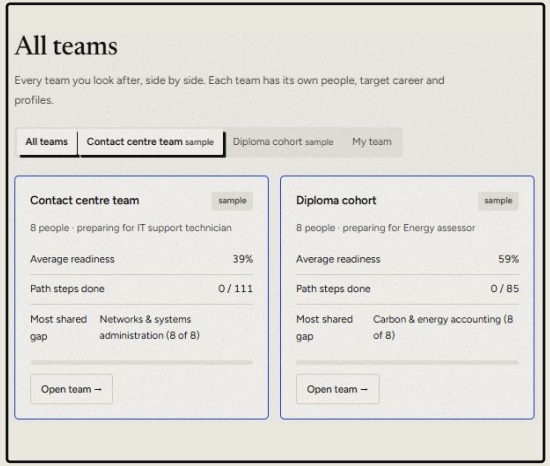
3 Get matched



4 Follow a pathway

FOR HR MANAGERS AND CAREER CENTRES

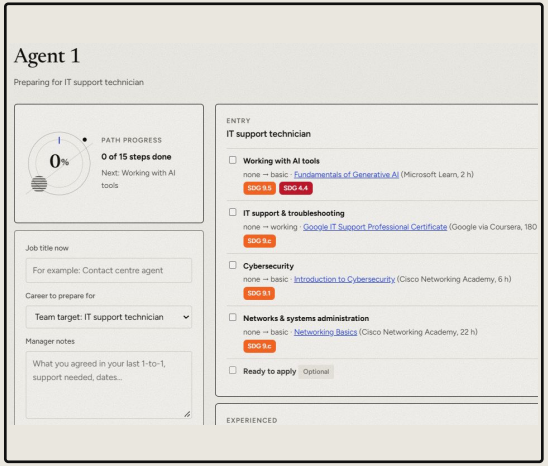
The same map for a whole team.



Several teams, each with its own target, compared on one page



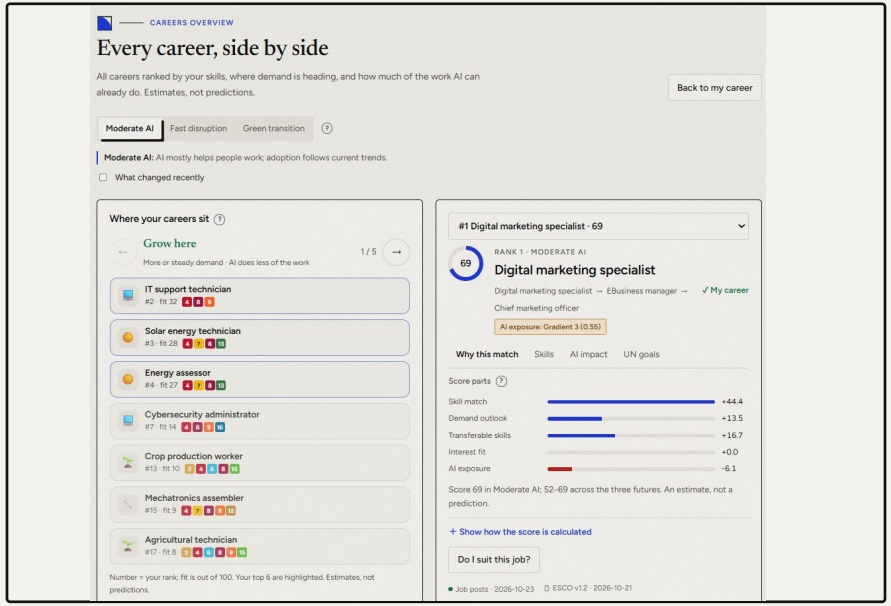
One skills table showing who meets each need



Each person's path, ticked off by the manager

03 · TECHNICAL APPROACH · TRACK AND PREDICT

Sensing the job market every day.



1 Collect · job posts and AI research, every day

2 Measure · each skill's share of posts against the 3 months before

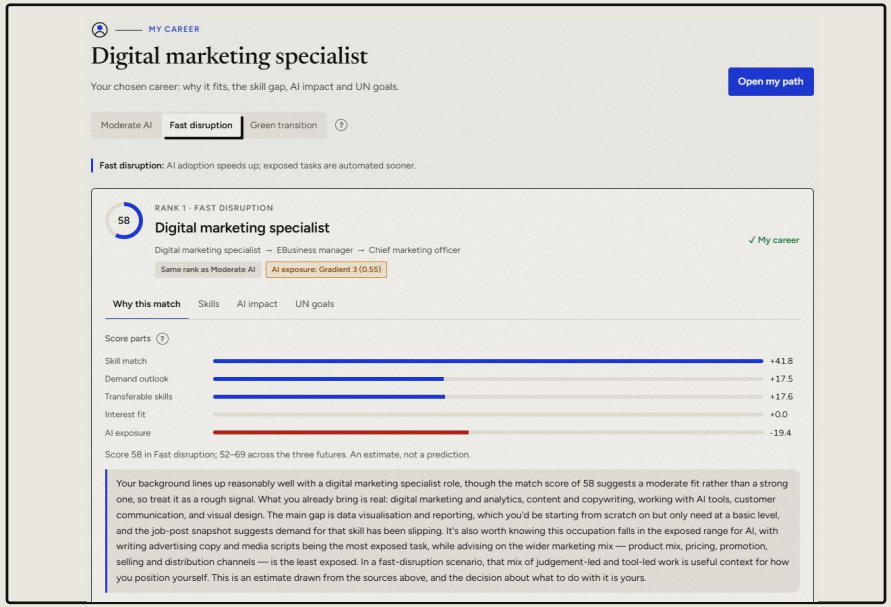
3 Spot · new tools such as MCP and AI agents

4 Remember · dated snapshots, compared at every update

Three futures, not one guess: every career is scored as a range.

Sources: Hacker News "Who is hiring?" and arXiv. Small samples are damped so a few posts can't swing a trend.

One skill language, one open formula.



People and jobs share one language: **ESCO v1.2** skills, in 39 clusters, across 20 careers.

\+ Skill match

\+ Demand outlook

\+ Transferable people skills

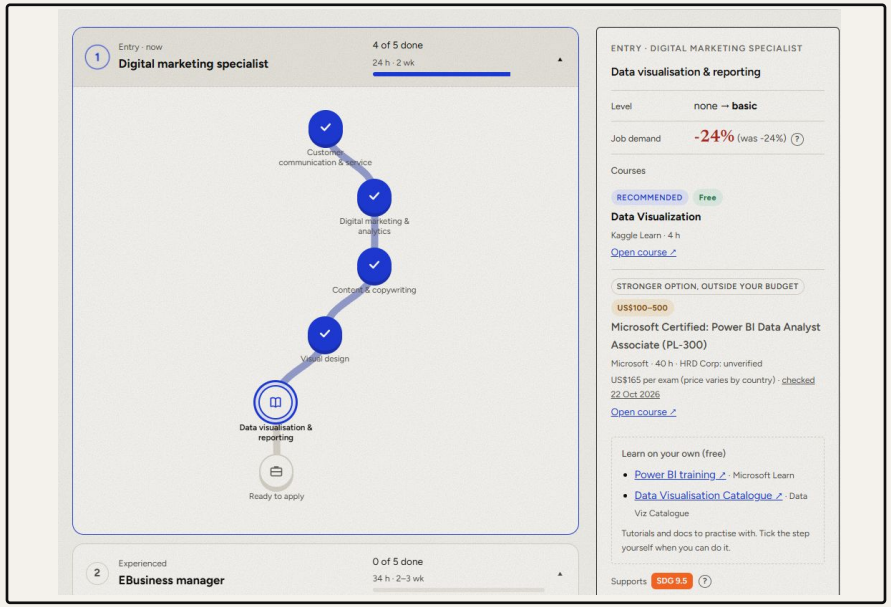
\+ Interest fit

\- AI exposure (ILO GenAI index)

= **Fit out of 100**, with the gap

Weights are published in the app. Skills ESCO predates, such as working with AI tools, are added and labelled.

A reason for every step.



- **Ordered** · prerequisites first, then what employers ask for most now
- **Learnable** · 71 checked courses, filtered by budget and weekly hours
- **Linked to the SDGs** · official UN targets, each with a reason
- **A real ladder** · three ESCO jobs, entry role to peak role

Links are tested and prices appear only when verified. SDG links don’t change rankings, except in the Green future.

LIVE DEMO · ABOUT 2 MINUTES

Let’s walk one person through it.

0:00	**Sign up and the story**	Tell us about yourself, then the four-step story with the person’s own numbers.
0:40	**Home and Path**	The next step, why this career fits, and a path with courses and SDG targets.
1:20	**Updates**	Press Update data: new job posts arrive and each section shows what changed.
1:50	**Team view**	All teams, then one member’s profile with path steps ticked by the manager.

Try it: https://workforce-future.replit.app/

03 · TECHNICAL APPROACH · TRUST

AI that explains, but never decides.

The language model may

- Read a CV or description and pick skills from a fixed list, quoting the text for each
- Reword an explanation from facts the engine has computed

It may not

- Choose careers, courses or scores
- Add a number that isn't in our data: that text is rejected and a template is shown
- See contact details: they are removed first

Fairness Check: age, gender and ethnicity never enter a score, and a test checks that age band can't change matches.

Stack: FastAPI · PostgreSQL · vanilla JS · DeepSeek via an OpenAI-compatible API · English, Bahasa Malaysia, Chinese

04 · EXPECTED IMPACT

From guessing to a plan people follow.

People

Know which skills to learn next and why, in their own language, with free or low-cost courses first.

Employers and campuses

Plan reskilling for whole teams and cohorts, train shared gaps once and track progress per person.

Malaysia and the SDGs

Steps toward critical occupations, with every skill and career linked to UN SDG targets.

SDG 4.4 SDG 8.5 SDG 8.6

WORKFORCE FUTURES

A career map that moves as fast as the job market.



Try it: https://workforce-future.replit.app/

Thank you · MK GreenCode